SC2104 :

SENSORS,
INTERFACING
AND
DIGITAL CONTROL

Part 1/II :

Analogue/Digital Filters

Lecturer SC2104 : –

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LEARNING OBJECTIVES :

Derive the Transfer Function of Analogue Low-Pass/High-Pass Filters. in the Time Domain, Frequency Domain, Laplace Transform Domain

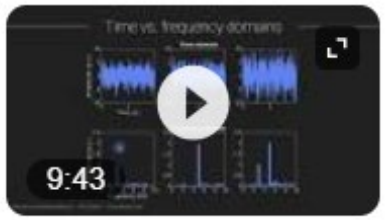
Design considerations when using an Analogue Filter.

Digital Filters – Introduction

Digital Filter Design Method (via Bilinear Transformation)

INTRO VIDEOS

Time and frequency domains - YouTube



Sine waves · Morlet wavelets in time and in frequency · Module 1: Time vs Frequency Domains · Control Systems Lectures - **Time and Frequency**…

YouTube · Mike X Cohen · 20 Dec 2019

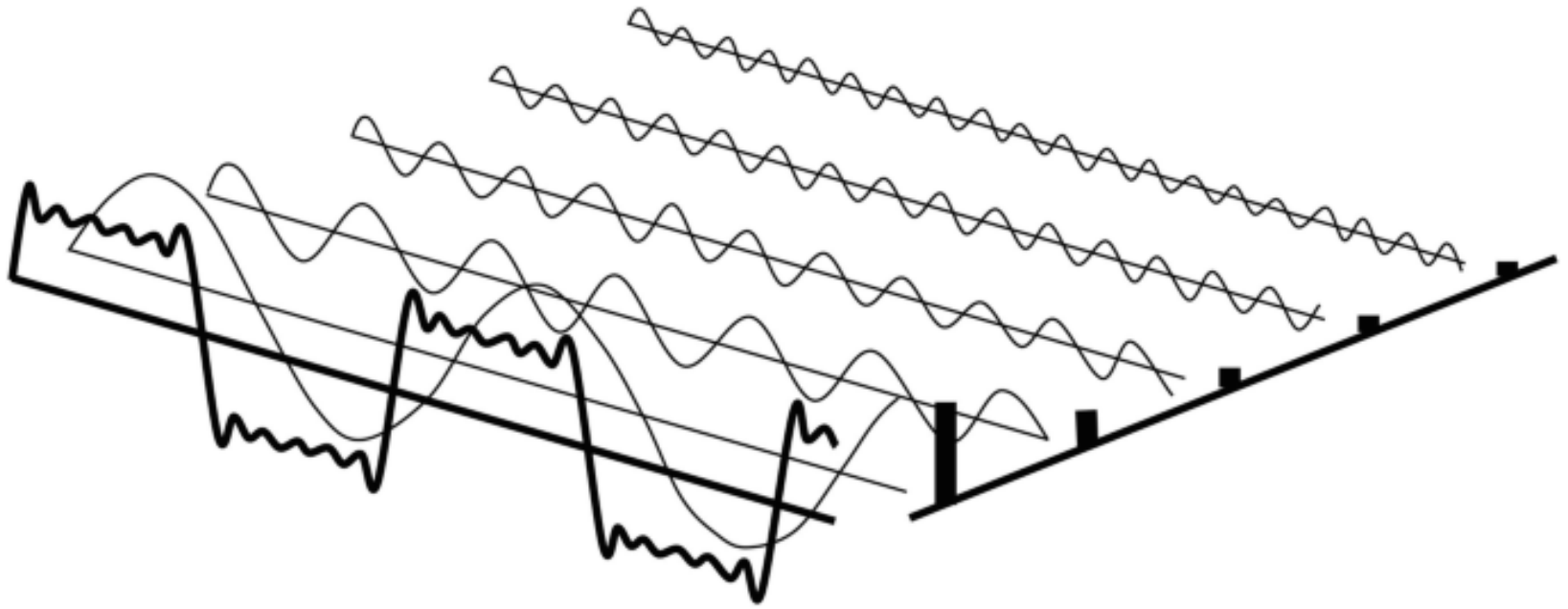
The Fourier Series and Fourier Transform Demystified



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YouTube · Up and Atom · 30 Jun 2022

FOURIER SERIES & FOURIER TRANSFORM (TOUCH & GO)



FOURIER TRANSFORM (TOUCH & GO)

Fourier Transform

The Fourier transform of a continuous-time function $x(t)$ can be defined as,

$$\mathrm{x}(\omega) = \int_{-\infty}^{\infty} \mathrm{x}(\mathrm{t}) \mathrm{e}^{-\mathrm{j}\omega \mathrm{t}} \mathrm{dt}$$

Fourier Transform of Sine Function

Let

$$\mathrm{x}(\mathrm{t}) = \sin \omega_0 \mathrm{t}$$

From Euler's rule, we have,

$$\mathrm{x}(\mathrm{t}) = \sin \omega_0 \mathrm{t} = \left[\frac{\mathrm{e}^{\mathrm{j}\omega_0 \mathrm{t}} - \mathrm{e}^{-\mathrm{j}\omega_0 \mathrm{t}}}{2\mathrm{j}} \right]$$

Then, from the definition of Fourier transform, we have,

$$\mathrm{F}[\sin \omega_0 \mathrm{t}] = \mathrm{X}(\omega) = \int_{-\infty}^{\infty} \mathrm{x}(\mathrm{t}) \mathrm{e}^{-\mathrm{j}\omega \mathrm{t}} \mathrm{dt} = \int_{-\infty}^{\infty} \sin \omega_0 \, \mathrm{t} \, \mathrm{e}^{-\mathrm{j}\omega \mathrm{t}} \mathrm{dt}$$

$$\Rightarrow \mathrm{X}(\omega) = \int_{-\infty}^{\infty} \left[\frac{\mathrm{e}^{\mathrm{j}\omega_0 \mathrm{t}} - \mathrm{e}^{-\mathrm{j}\omega_0 \mathrm{t}}}{2\mathrm{j}} \right] \mathrm{e}^{-\mathrm{j}\omega \mathrm{t}} \mathrm{dt}$$

$$\Rightarrow \mathrm{X}(\omega) = \frac{1}{2\mathrm{j}} \left[\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{j}\omega_0 \mathrm{t}} \mathrm{e}^{-\mathrm{j}\omega \mathrm{t}} \mathrm{dt} - \int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{j}\omega_0 \mathrm{t}} \mathrm{e}^{-\mathrm{j}\omega \mathrm{t}} \mathrm{dt} \right]$$

$$= \frac{1}{2\mathrm{j}} \{ \mathrm{F}[\mathrm{e}^{\mathrm{j}\omega_0 \mathrm{t}}] - \mathrm{F}[\mathrm{e}^{-\mathrm{j}\omega_0 \mathrm{t}}] \}$$

Since, the Fourier transform of complex exponential function is given by,

$$\mathrm{F}[\mathrm{e}^{\mathrm{j}\omega_0 \mathrm{t}}] = 2\pi\delta(\omega - \omega_0) \text{ and } \mathrm{F}[\mathrm{e}^{-\mathrm{j}\omega_0 \mathrm{t}}] = 2\pi\delta(\omega + \omega_0)$$

$$\therefore \mathrm{X}(\omega) = \frac{1}{2\mathrm{j}} [2\pi\delta(\omega - \omega_0) - 2\pi\delta(\omega + \omega_0)]$$

$$\Rightarrow \mathrm{X}(\omega) = -\mathrm{j}\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

Therefore, the Fourier transform of the sine wave is,

$$\mathrm{F}[\sin \omega_0 \, \mathrm{t}] = -\mathrm{j}\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

Or, it can also be represented as,

$$\sin \omega_0 \, \mathrm{t} \overset{\mathrm{FT}}{\leftrightarrow} -\mathrm{j}\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

The graphical representation of the sine function with its magnitude and phase spectra is shown in Figure-1.

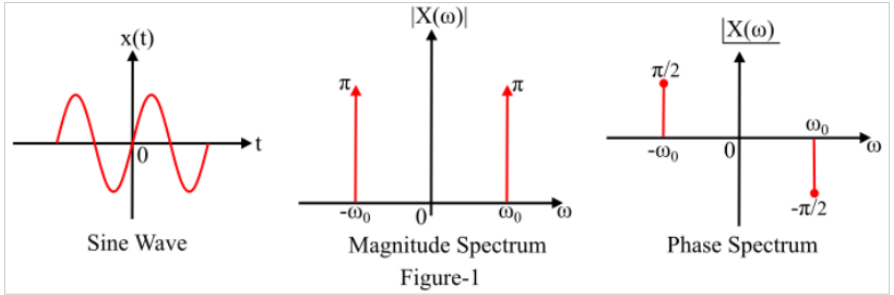


Figure-1

LAPLACE TRANSFORM (TOUCH & GO)

Fourier Transform

The Fourier transform of a continuous-time function $x(t)$ can be defined as,

$$x(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Laplace Transform

$$F(s) = \int_{0}^{\infty} f(t) e^{-st}\, dt$$

where $s = \sigma + i\omega$

$F(s)$ = laplace transform

s = complex number

t = real number >= 0

One-sided Integration & the expontential term consist of real & imaginary part

LAPLACE TRANSFORM (TOUCH & GO)

Laplace Trans

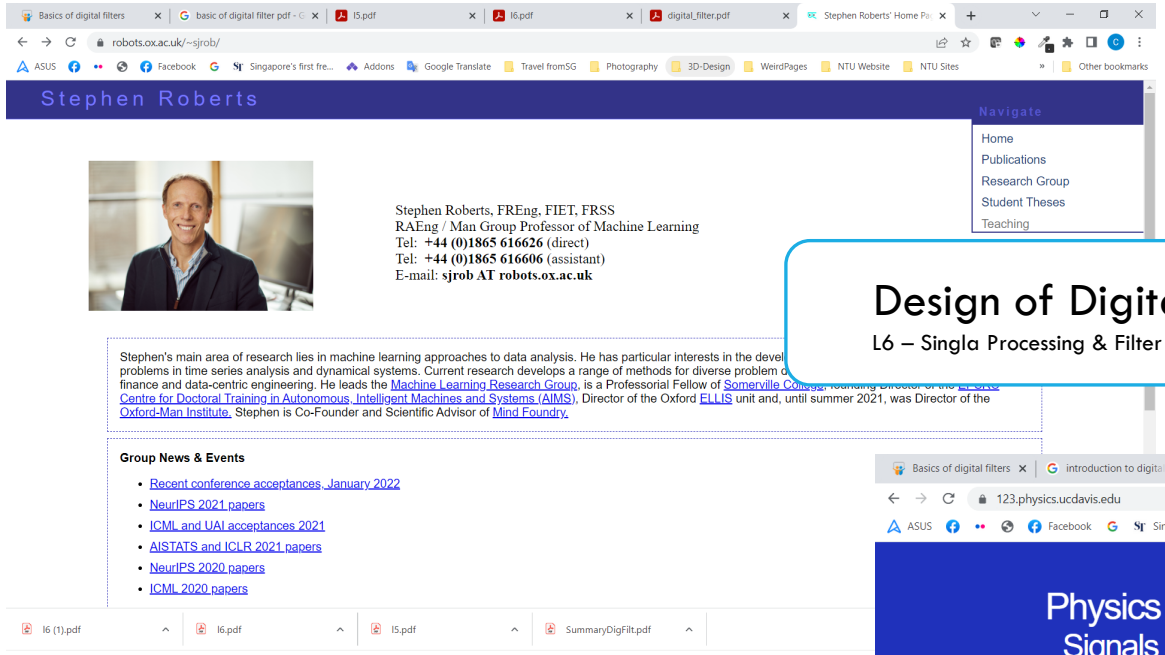
$F(s)$ = laplace trar

s = complex nu

t = real numbe

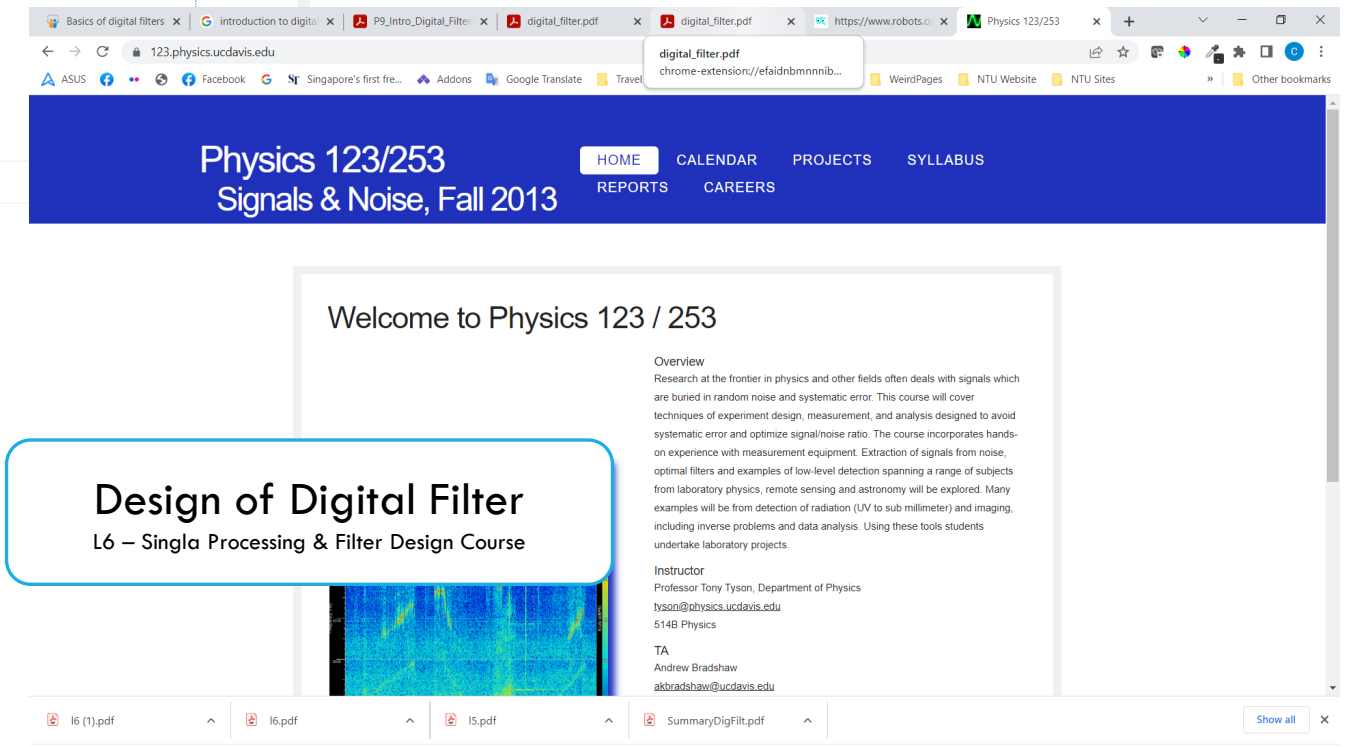
S.no	$f(t)$	$\mathcal{L}\{f(t)\}$	S.no	$f(t)$	$\mathcal{L}\{f(t)\}$
1	1	$\frac{1}{s}$	11	$e^{at}\sinh bt$	$\frac{b}{(s-a)^2-b^2}$
2	e^{at}	$\frac{1}{s-a}$	12	$e^{at}\cosh bt$	$\frac{s-a}{(s-a)^2-b^2}$
3	t^n	$\frac{n!}{s^{n+1}}$	13	$t\cos at$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
4	$\sin at$	$\frac{a}{s^2+a^2}$	14	$t\sin at$	$\frac{2as}{(s^2+a^2)^2}$
5	$\cos at$	$\frac{s}{s^2+a^2}$	15	$f'(t)$	$sF(s)-f(0)$
6	$\sinh at$	$\frac{a}{s^2-a^2}$	16	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
7	$\cosh at$	$\frac{s}{s^2-a^2}$	17	$\int_0^t f(u)du$	$\frac{1}{s}F(s)$
8	$e^{at}t^n$	$\frac{n!}{(s-a)^{n+1}}$	18	$t^nf(t)$ Where $n=1,2,3,..$	$(-1)^n\frac{d^n}{ds^n}\{F(s)\}$
9	$e^{at}\cos bt$	$\frac{s-a}{(s-a)^2+b^2}$	19	$\frac{1}{t}\{f(t)\}$	$\int_s^\infty F(s)ds$
10	$e^{at}\sin bt$	$\frac{b}{(s-a)^2+b^2}$	20	$e^{at}f(t)$	$F(s-a)$

ANALOGUE-DIGITAL FILTERS



Design of Digital Filter

L6 – Singla Processing & Filter Design Course

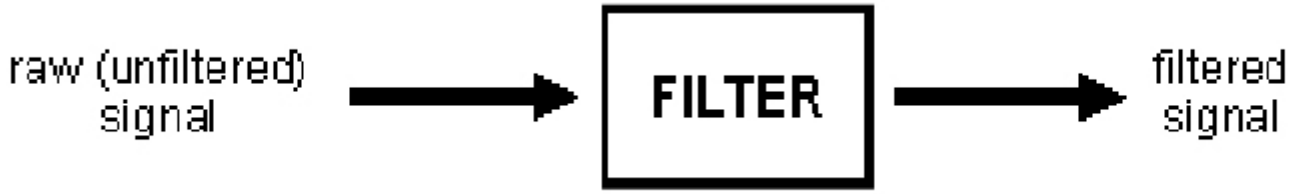


Design of Digital Filter

L6 – Singla Processing & Filter Design Course

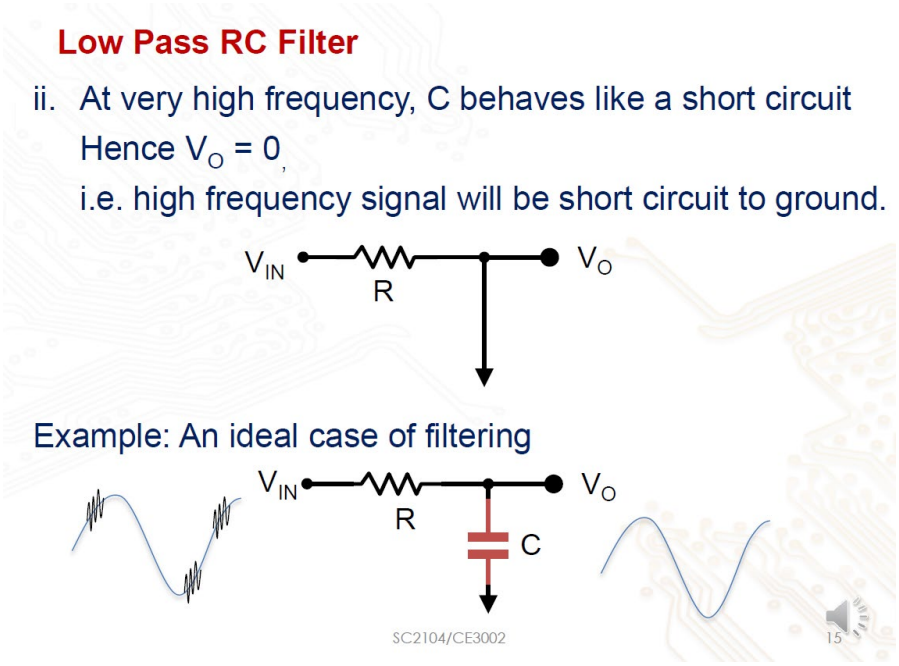
FILTER (ANALOG/DIGITAL)

In signal processing, the function of a filter is to remove unwanted parts of the signal, such as random noise, or to extract useful parts of the signal, such as the components lying within a certain frequency range.



Filters are used in applications like data compression, biomedical signal processing, speech and image processing, data transmission, digital audio, telephone echo cancellation etc

WATCH A VIDEO



Listen to example of Low-Pass Filter & High-Pass Filter in a Audio/Music Clip

[Audio and Music Tutorial - EQ with high and low pass filters]

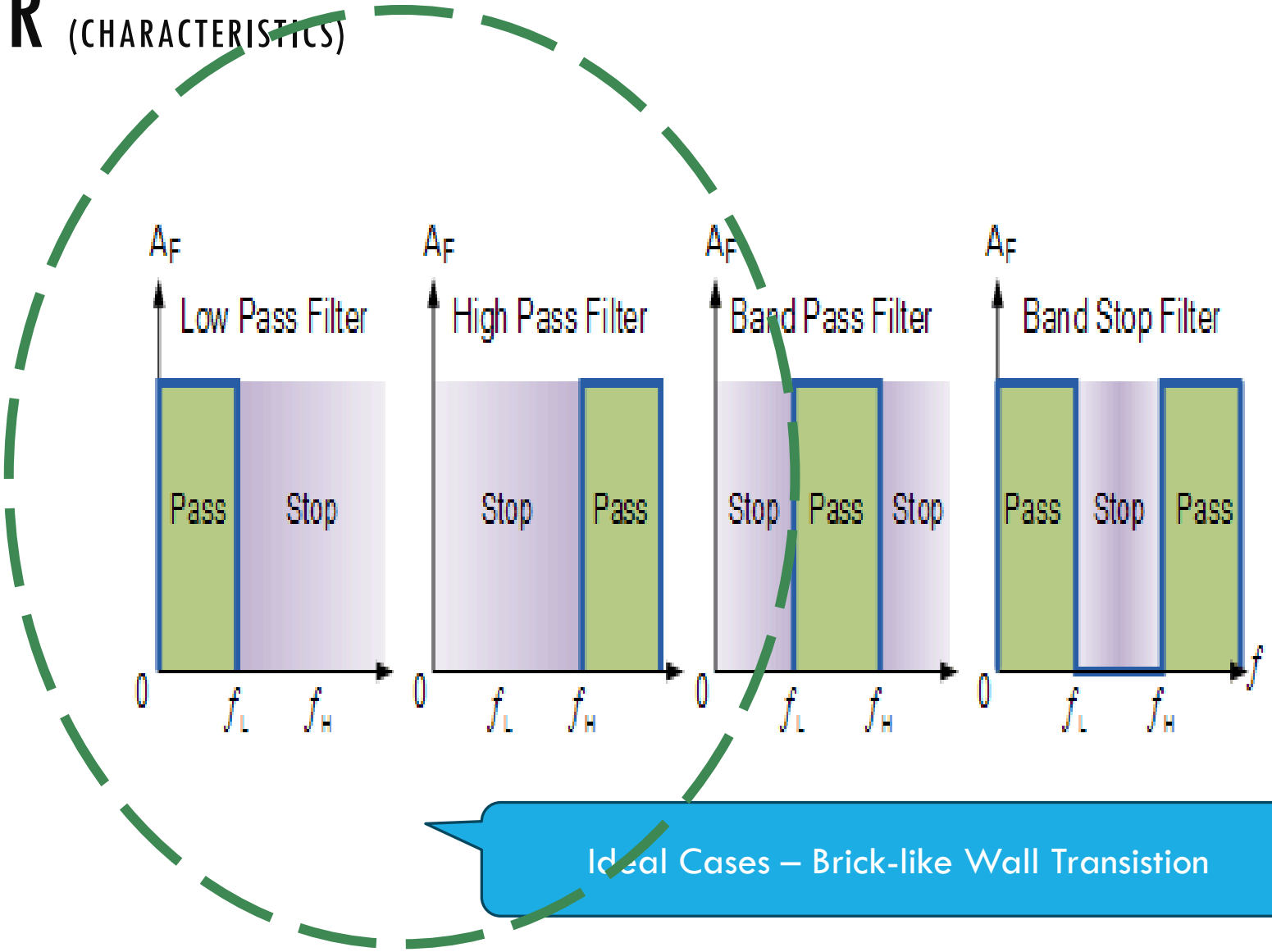
FILTER (ANALOG/DIGITAL)

There are two main kinds of filter, analog and digital. They are quite different in their physical makeup and in how they work.

An analogue filter uses analogue electronic circuits made up from components such as resistors, capacitors and op amps to produce the required filtering effect. Such filter circuits are widely used in such applications as noise reduction, video signal enhancement, graphic equalisers in hi-fi systems, and many other areas. There are well-established standard techniques for designing an analogue filter circuit for a given requirement. At all stages, the signal being filtered is an electrical voltage or current which is the direct analogue of the physical quantity (e.g. a sound or video signal or transducer output) involved.

TYPES OF FILTER (CHARACTERISTICS)

1. Low Pass Filter
2. High Pass Filter
3. Band Pass Filter
4. Band Stop Filter

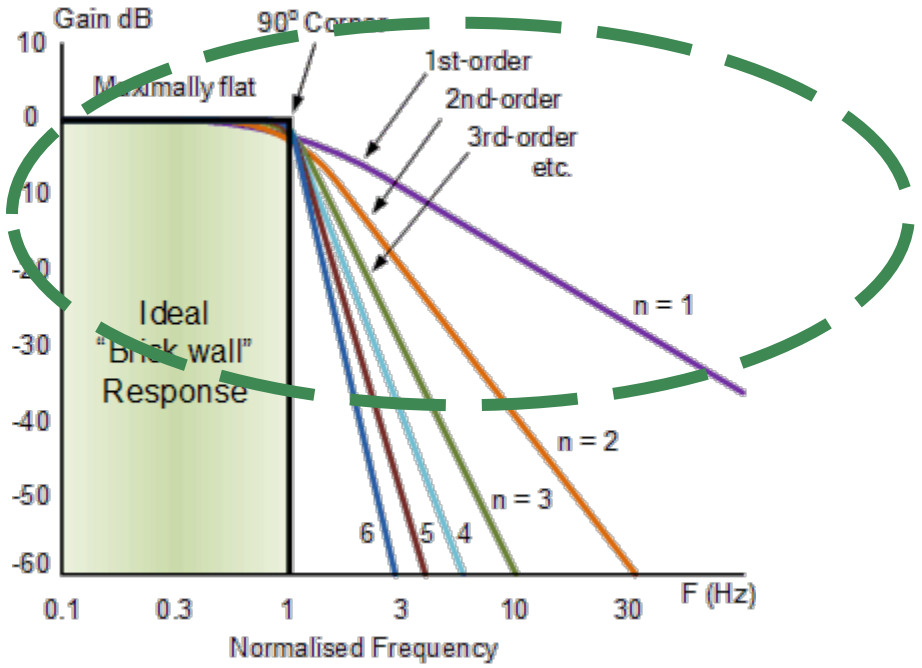


FILTER CHARACTERISTICS

The order n of an Analogue Filter is related to the number of reactive elements (Capacitors & Inductors) in the circuit.

It implies that a n-order filter will have an attenuation rate of 20dB/decade.

Passive Filter has a Passband Gain of 1 whereas an Active Filter can have Passband Gain > 1 (where op-amp circuit is used to give the required amplification)



LOW-PASS FILTER (ANALOGUE)

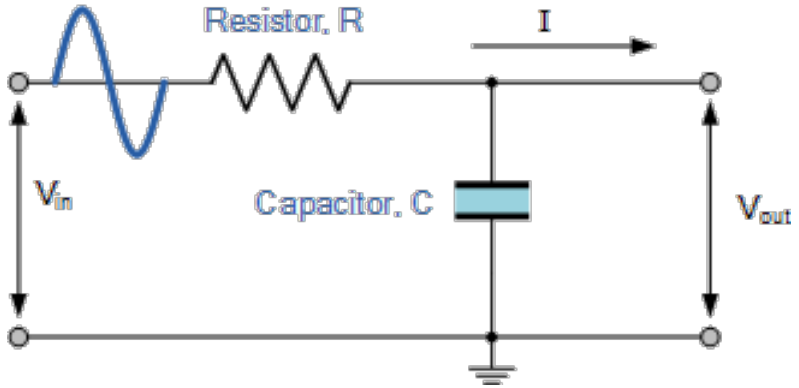
Voltage Division Rule :

$$V_o = \left(\frac{X_c}{R + X_c}\right) V_i$$

$$where \quad X_c = \frac{1}{j2\pi fC}$$

$$V_o(f) = \left(\frac{\frac{1}{j2\pi fC}}{R + \frac{1}{j2\pi fC}}\right) V_i(f)$$

$$V_o(f) = \left(\frac{\frac{1}{j2\pi f_c}}{\frac{1 + j2\pi fRC}{j2\pi f_c}}\right) V_i(f)$$

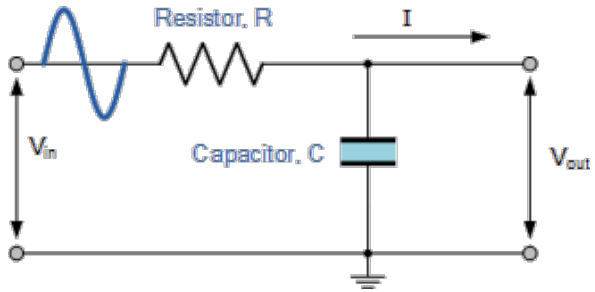


RC Network (Low Pass Filter)

Transfer Function in Freq Domain

$$\frac{V_o(f)}{V_i(f)} = \left(\frac{1}{1 + j2\pi fRC}\right)$$

LOW-PASS FILTER (ANALOGUE)



RC Network (Low Pass Filter)

$$\frac{V_o}{V_i} = \left(\frac{1}{1 + j2\pi fRC}\right) \quad let \quad 2\pi f_c = \frac{1}{RC}$$

Corner Freq

$$\frac{V_o}{V_i} = \left(\frac{1}{1 + j(2\pi f / 2\pi f_c)}\right)$$

$$\frac{V_o}{V_i} = \left(\frac{1}{1 + j\, f / f_c}\right)$$

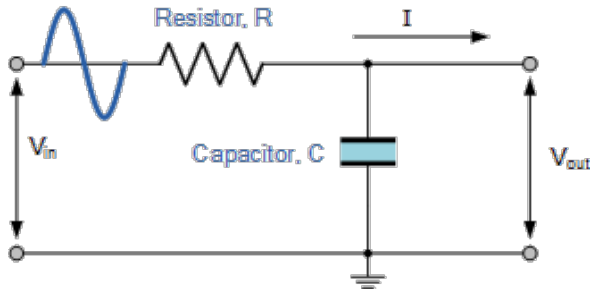
Generic Transfer Function of a Low-Pass Filter

$$\left|\frac{V_o}{V_i}\right| = \left(\frac{1}{\sqrt{1^2 + (f/f_c)^2}}\right)$$

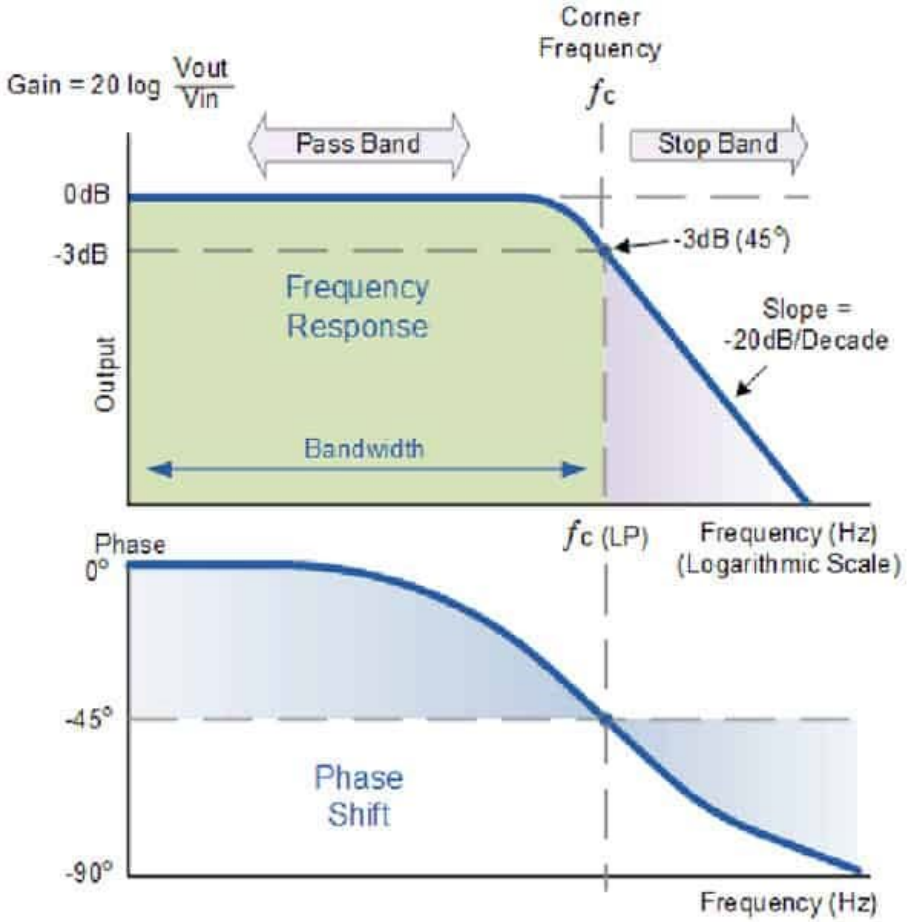
You can convert this transfer function from the rectangular equation into its polar equation. And for our analysis, we will just consider the magnitude portion and put aside the phase portion of the polar equation.

LOW-PASS FILTER (ANALOGUE)

$$\left|\frac{V_o}{V_i}\right| = \left(\frac{1}{\sqrt{1^2 + (f/f_c)^2}}\right)$$



RC Network (Low Pass Filter)



LOW-PASS FILTER (ANALOGUE)

Another Derivation

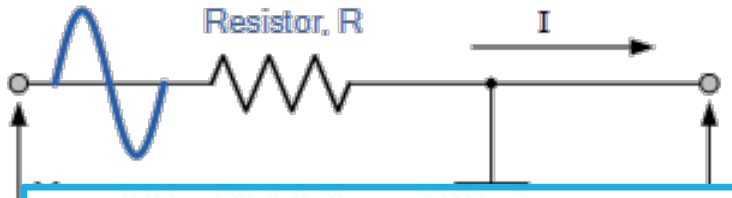
Kirchhoff's voltage law :

$$V_i = V_R + V_c$$

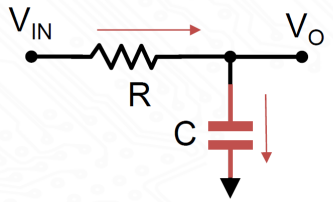
$$V_i(t) = i(t)R + V_o(t)$$

$$V_i(t) = RC\frac{dV_o(t)}{dt} + V_o(t)$$

Low-Pass Filter Function in Time Domain



Low Pass Filtering



Current flowing through R:
$I_R = (V_{IN} - V_O)/R$

Current flowing through C:
$I_C = C\, dV_O/dt$

$I_R = I_C \quad \Rightarrow \quad (V_{IN} - V_O)/R = C\, dV_O/dt$

Rearranging $V_O + RC\, dV_O/dt = V_{IN}$ ------ (1)

Implement differentiation digitally in software:

$dV_O/dt = (V_O[n] - V_O[n-1]) / T$ ------ (2)

where [n] is the current value, [n-1] is the pervious value and T is the time in between the two values

LOW-PASS FILTER (ANALOGUE)

Kirchhoff's voltage law :

$$V_i = V_R + V_c$$

$$V_i(t) = i(t)R + V_o(t)$$

$$V_i(t) = RC\frac{dV_o(t)}{dt} + V_o(t)$$

Low-Pass Filter Function in Time Domain

$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = sF(s) - f(0)$$

$$\mathcal{L}\left\{\frac{d^2}{dt^2}f(t)\right\} = s^2F(s) - sf(0) - f'(0)$$

$$\mathcal{L}\left\{\frac{d^3}{dt^3}f(t)\right\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0)$$

etc.

Transfer Function in Laplace Transform Domain

LPF TRANSFER FUNCTIONS (COMPARING)

LPF Transfer Function (Method 1)

$$H_{LPF}(s) = \frac{V_o(f)}{V_i(f)} = \left(\frac{1}{1 + j2\pi fRC}\right) = \left(\frac{{}^{1}/_{RC}}{j2\pi f + {}^{1}/_{RC}}\right) = \left(\frac{{}^{1}/_{RC}}{j\omega + {}^{1}/_{RC}}\right) = \frac{{}^{1}/_{RC}}{s + {}^{1}/_{RC}}$$



LPF Transfer Function (Method 2)

$$H_{LPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{{}^{1}/_{RC}}{s + {}^{1}/_{RC}}$$

Let

$$s = j\omega$$

TRANSFER FUNCTIONS (EXTEND FURTHER)

Corner Frequency :

$$2\pi f_c = \frac{1}{RC} \quad \text{or} \quad \omega_c = \frac{1}{RC}$$

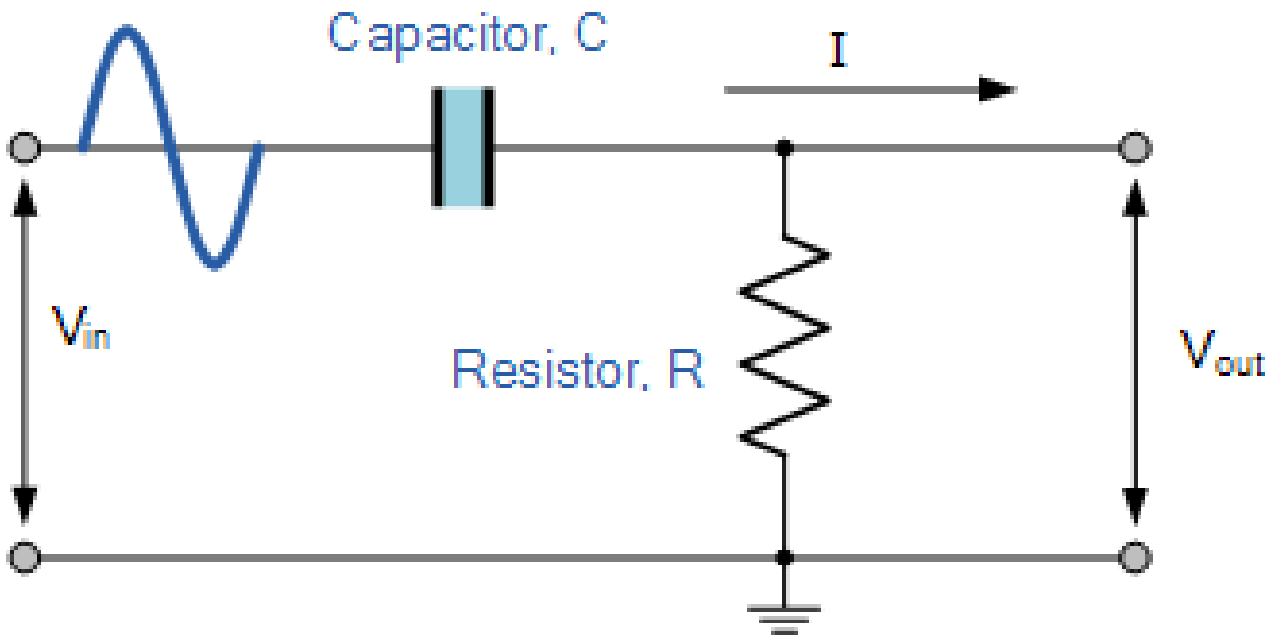
Generic LPF Transfer Function (Method 1)

$$H_{LPF}(s) = \frac{V_o(f)}{V_i(f)} = \left(\frac{1}{1 + j2\pi fRC}\right) = \left(\frac{{}^{1}/_{RC}}{j2\pi f + {}^{1}/_{RC}}\right) = \left(\frac{\omega_c}{j\omega + \omega_c}\right) = \left(\frac{\omega_c}{s + \omega_c}\right)$$

Generic LPF Transfer Function (Method 2)

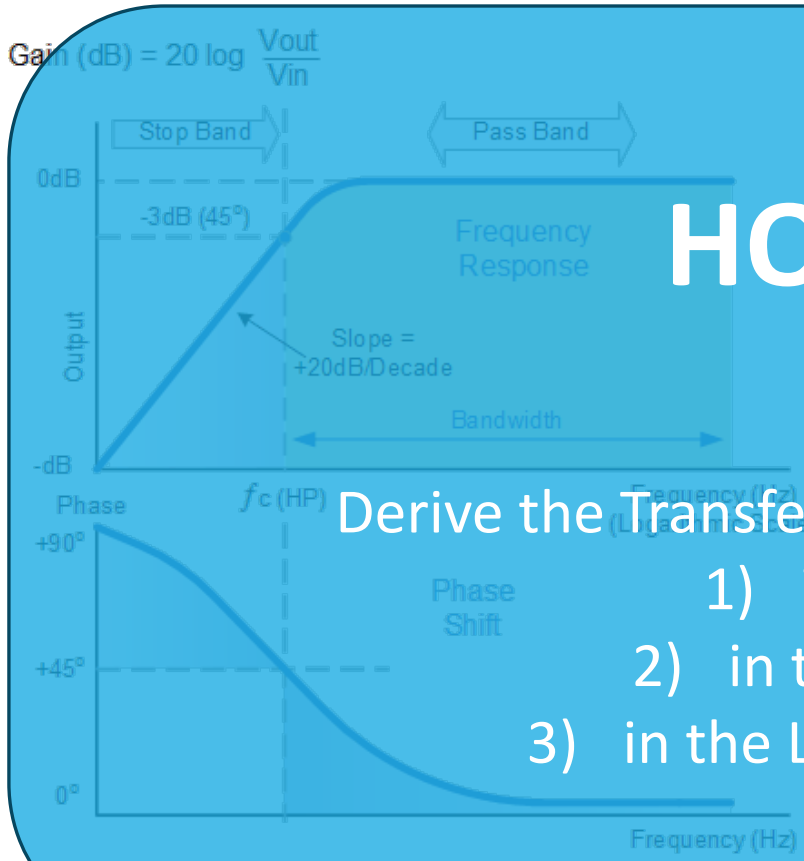
$$H_{LPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{{}^{1}/_{RC}}{s + {}^{1}/_{RC}} = \frac{\omega_c}{s + \omega_c}$$

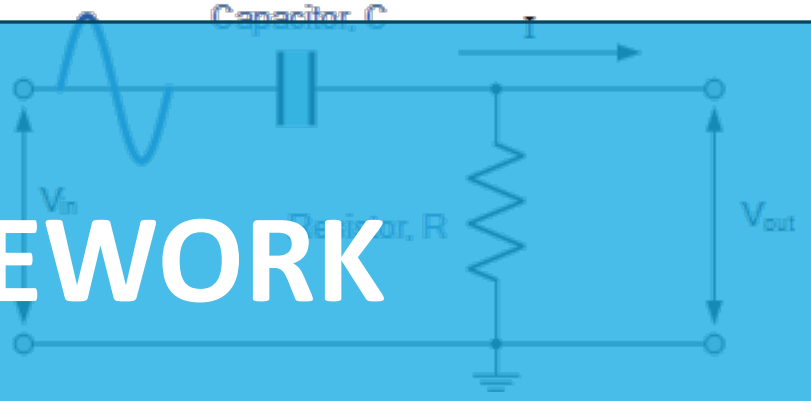
HIGH-PASS FILTER (ANALOGUE)



RC Network (High Pass Filter)

HIGH-PASS FILTER (ANALOGUE)





RC Network (High Pass Filter)

HOMEWORK

Derive the Transfer Function of the High Pass Filter :

1) in the time domain,
2) in the frequency domain,
3) in the Laplace Transform domain.

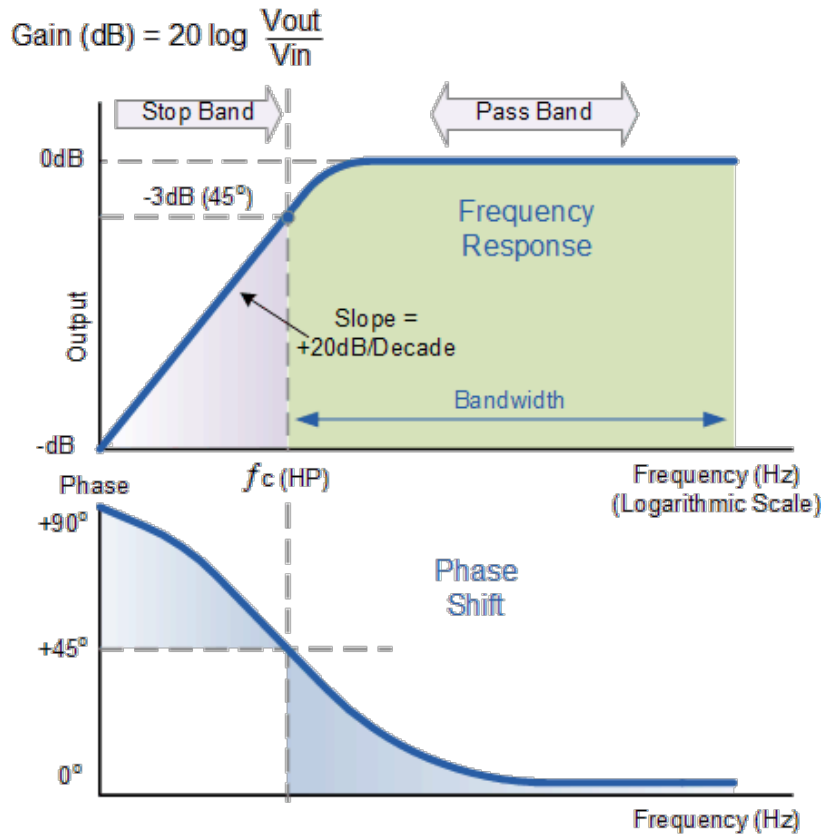
$$\frac{V_o(f)}{V_i(f)} = \left(\frac{j2\pi fRC}{1 + j2\pi fRC}\right)$$

Freq Domain

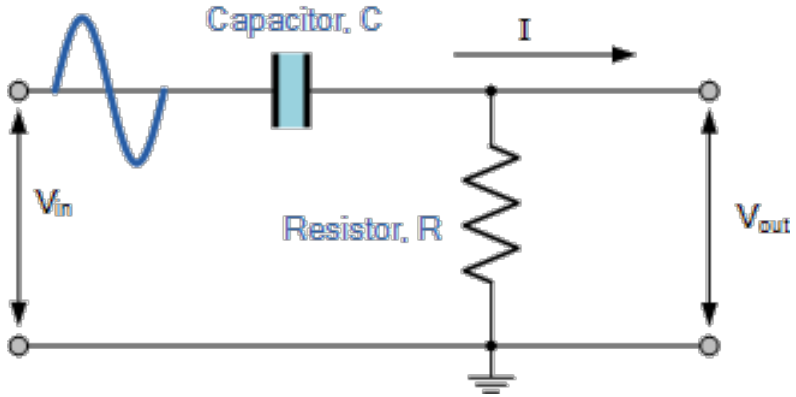
$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{s}{s + {}^{1}/_{RC}}$$

Laplace Transform Domain

HIGH-PASS FILTER (ANALOGUE)



Laplace Transform Domain



RC Network (High Pass Filter)

$$\frac{V_o(f)}{V_i(f)} = \left(\frac{j2\pi fRC}{1+j2\pi fRC}\right)$$

Freq Domain

$$H_{HPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{s}{s+{}^{1}/_{RC}} = \frac{s}{s+\omega_c}$$

REMINDER :

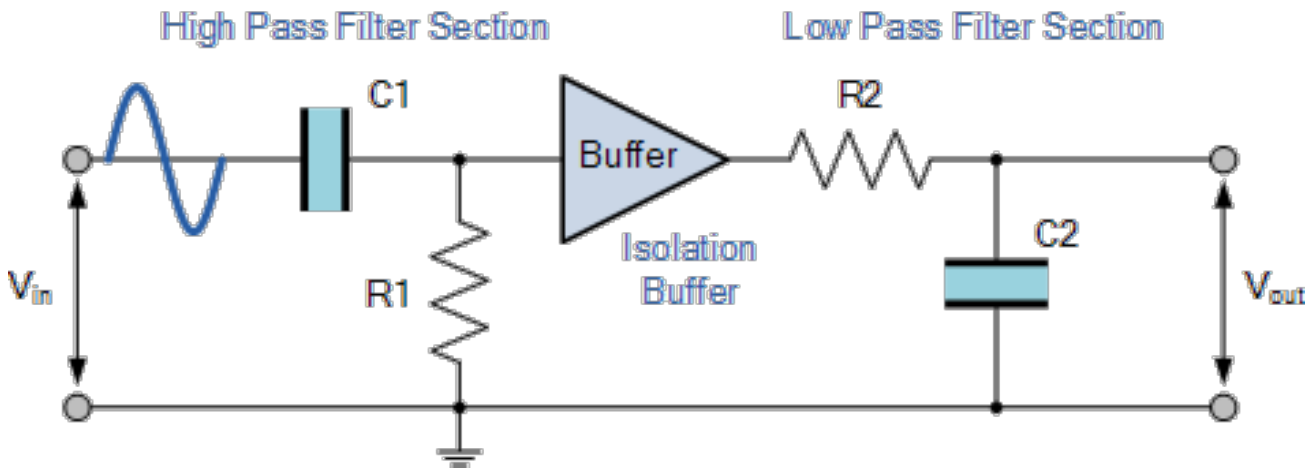
Generic LPF Transfer Function (Laplace Transform Domain)

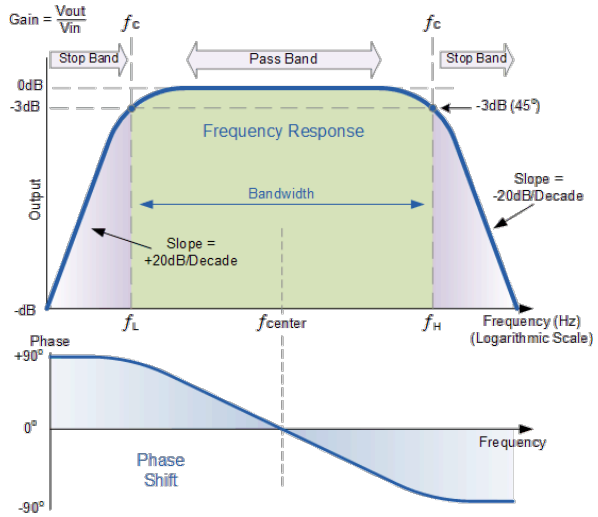
$$H_{LPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{\omega_c}{s + \omega_c}$$

Generic HPF Transfer Function (Laplace Transform Domain)

$$H_{HPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{s}{s + \omega_c}$$

BAND-PASS FILTER (ANALOGUE)

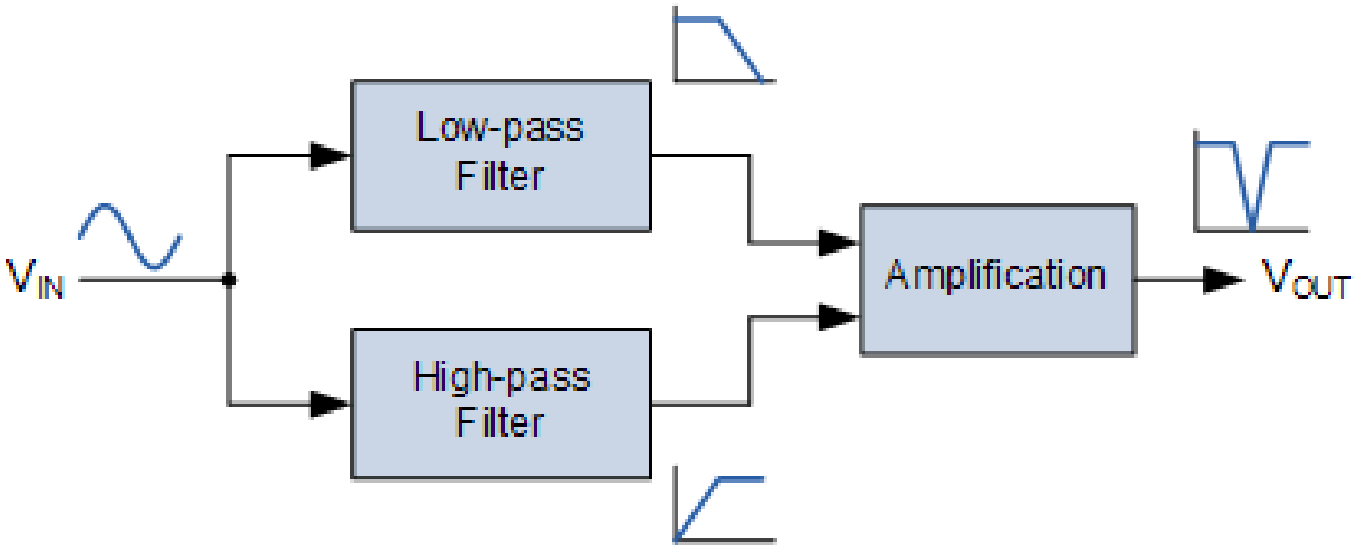


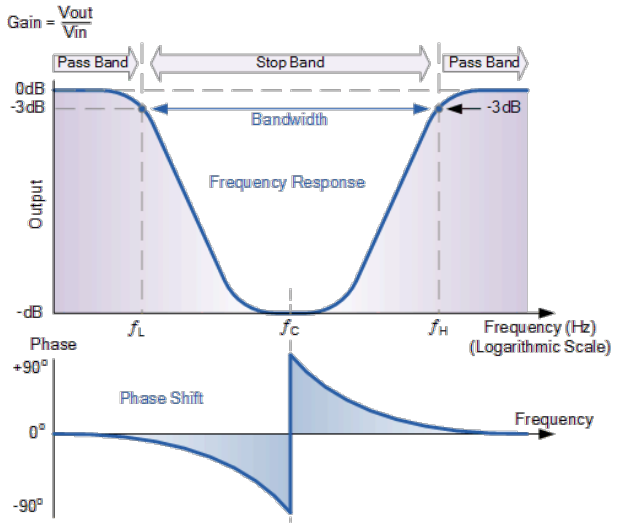


$f_{HC} < f_{LC}$

Cascade(Series) High-Pass & Low Pass Filters

BAND-STOP FILTER (ANALOGUE)





$f_{HC} > f_{LC}$

High-Pass & Low Pass Filters must be in parallel

FILTERS

Passive & Active Filters

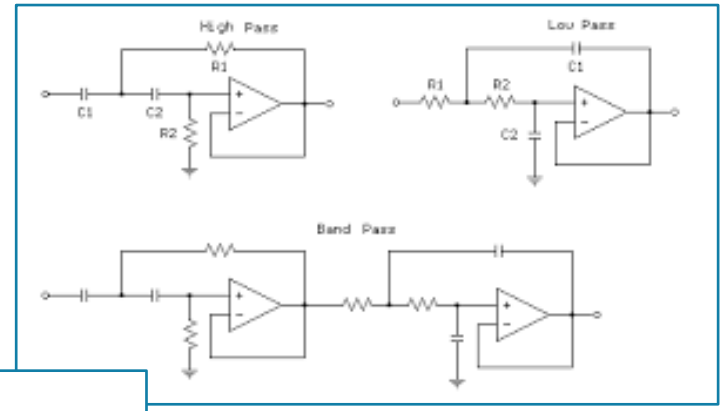
- Passive Filter has gain of 1
- Active Filters has Amplification (OpAmp)

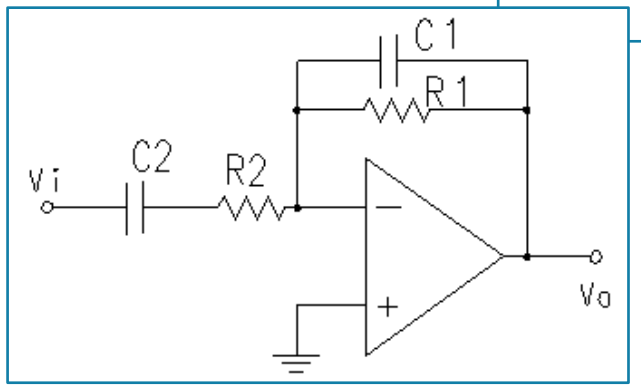
Higher Order Filters

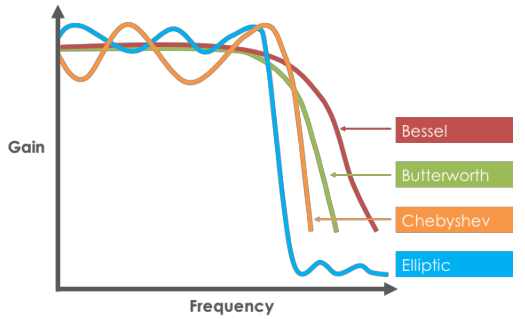
- Dropoff rate-20n DB/dec (-6n DB/octive)
- Require smaller freq transition range

Filters with Special Characteristics

- Butterworth Filter
- Chebyshev Filter
- Bessel Filter







DESIGN CONSIDERATIONS WHEN USING AN ANALOGUE FILTER

- Understand the Signal & Noise Spectrum
- Work out the Frequency Transition Range
- Decide on the Type and Order of the Filter
- Implement the Filter

EX DESIGN CONSIDERATIONS

A cable which carries sensor signal in the frequency range 40 kHz-60 kHz is corrupted with noise that has frequencies in the range 0-1 kHz. A filter circuit needs to be designed to reduce the noise level. The desired specifications of the filter are pass-band gain of 0 dB (Gain of 1) and at least -20dB attenuation at 1 kHz.

a) Determine the cut-off frequency of the filter that meets the desired specifications.

b) Propose a suitable filter circuit (R & C values) to meet the design requirement.

EX DESIGN CONSIDERATIONS

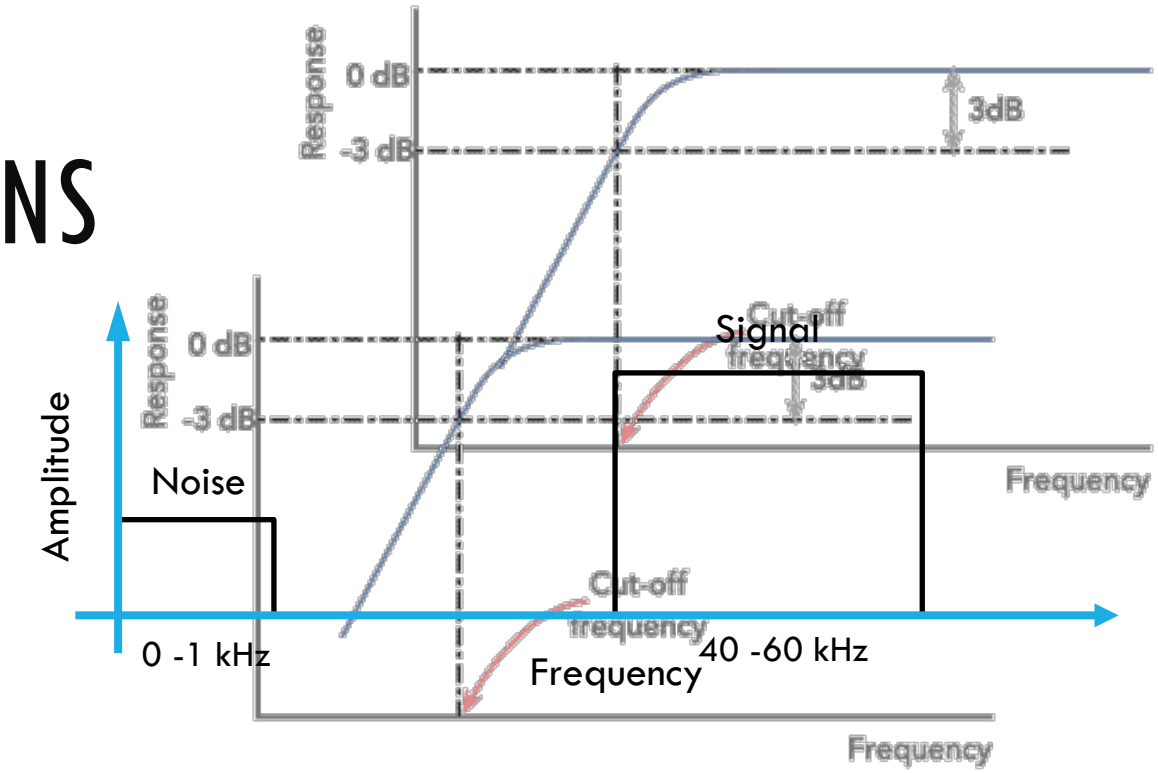
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EX DESIGN CONSIDERATIONS

- Understand the Signal & Noise Spectrum
- Work out the Frequency Transition Range
- Decide on the Type and Order of the Filter
- Implement the Filter



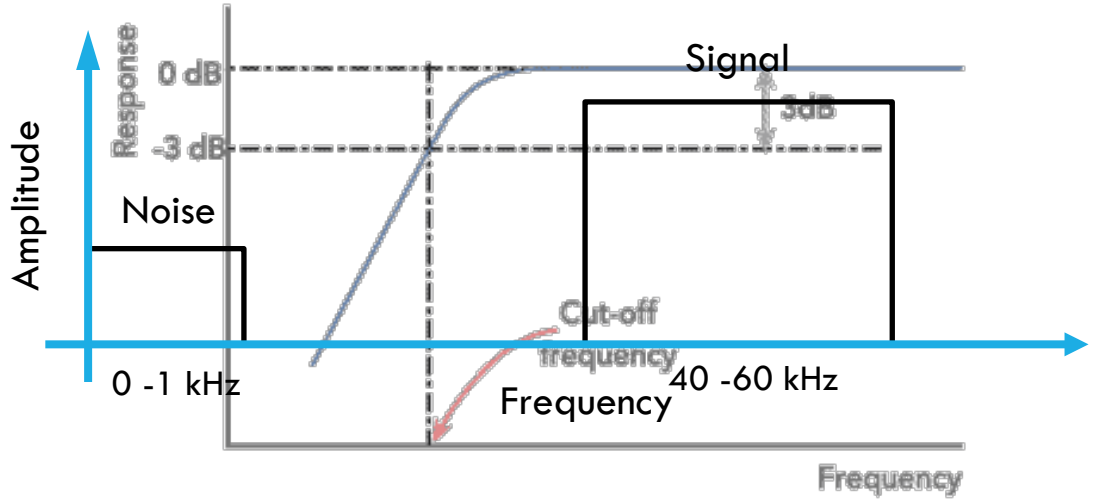
Maxi Noise Freq = 1KHz - Min Signal Freq = 40KHz

High-Pass Filter, Corner Frequency : 2-4 times lower than 40KHz, select f_c = 10KHz

Using 1st Order (-20dB/dec), the Noise, 1 decade away, will be attenuated by at least -20dB.

EX DESIGN CONSIDERATIONS

- Understand the Signal & Noise Spectrum
- Work out the Frequency Transition Range
- Decide on the Type and Order of the Filter
- Implement the Filter



$$\frac{V_o(f)}{V_i(f)} = \left(\frac{j2\pi fRC}{1 + j2\pi fRC}\right)$$

$$let\ 2\pi f_c = \frac{1}{RC}$$

$$\left|\frac{V_o}{V_i}\right| = \left(\frac{f/f_c}{\sqrt{1^2 + (f/f_c)^2}}\right)$$

FILTER (ANALOG/DIGITAL)

A **digital** filter uses a digital processor to perform numerical calculations on sampled values of the signal. The processor may be a general-purpose computer such as a PC, or a specialised DSP (Digital Signal Processor) chip.

The analogue input signal must first be sampled and digitised using an ADC (analogue to digital converter). The resulting binary numbers, representing successive sampled values of the input signal, are transferred to the processor, which carries out numerical calculations on them. These calculations typically involve multiplying the input values by constants and adding the products together. If necessary, the results of these calculations, which now represent sampled values of the filtered signal, are output through a DAC (digital to analogue converter) to convert the signal back to analogue form. Note that in a digital filter, the signal is represented by a sequence of numbers, rather than a voltage or current.

DIGITAL FILTER (TRANSFER

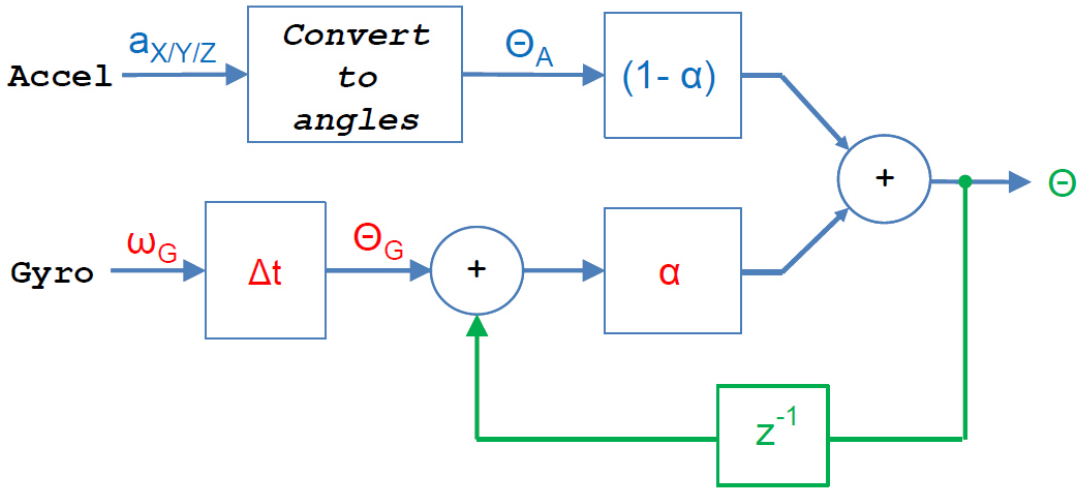
In the last few slides, we used two dif
filter: a form giving the output y_n dir
output terms on one side and all the i

Here, we introduce what is called the
obtained from the symmetrical form
describe a filter by means of a conve
transfer function of a filter to work ou

Discrete Time Complementary Filter Implementation

Complementary filter can be implemented in discrete time domain using the general equation

$$\Theta[n] = (1-\alpha) * (\Theta_A[n]) + \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t)$$



First of all, we must introduce the delay operator, denoted by the symbol z^{-1}. When applied to a sequence of digital values, this operator gives the previous value in the sequence. It therefore in effect introduces a delay of one sampling interval.

TYPES OF DIGITAL FILTERS

Infinite Impulse Response (IIR) Filters
(also known as recursive FILTER)

Finite Impulse Response (FIR) Filters
(also known as non-recursive FILTER)

INFINITE IMPULSE RESPONSE (IIR) FILTERS

(ALSO KNOWN AS RECURSIVE FILTER)

Generic Form :

$$H(z) = \frac{y_n(z)}{x_n(z)} = \frac{(b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots.. + b_M z^{-M})}{(a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots.. + a_N z^{-N})}$$

Transfer Function
(z-domain)

$$y(m) = \sum_{k=1}^{N} a_k \, y(m-k) + \sum_{k=0}^{M} b_k \, x(m-k)$$

Difference Equation
(discrete time samples)

also known as
Input – Output Equation

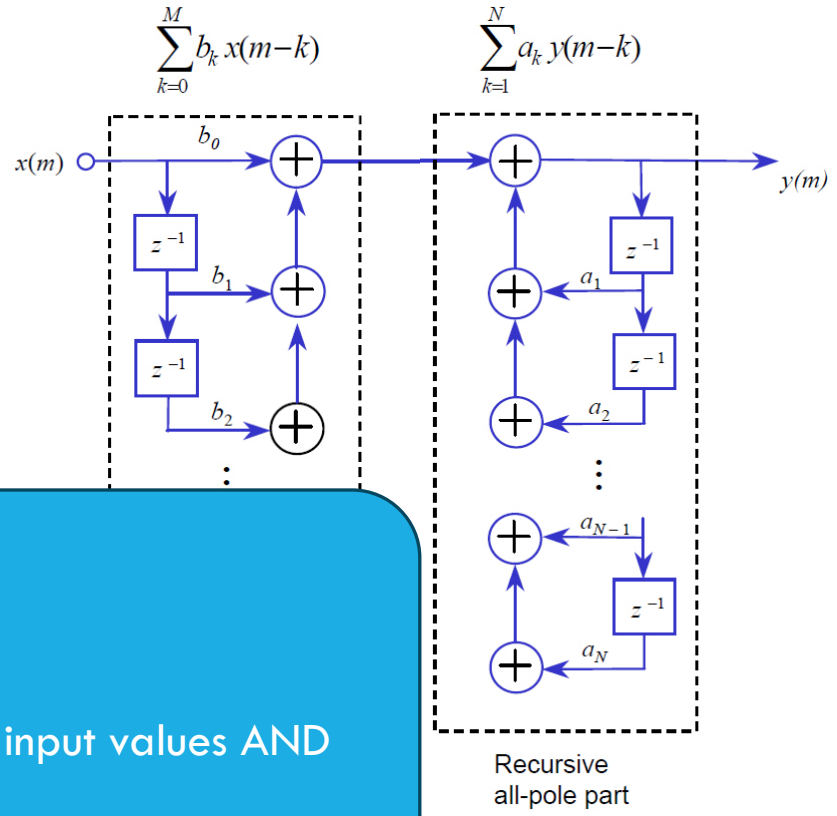
INFINITE IMPULSE RESPONSE (IIR) FILTERS

(ALSO KNOWN AS RECURSIVE FILTER)

A recursive filter has feedback from output to input, and in general its output is a function of the previous output samples and the present and past input samples as described by the following equation

$$y(m) = \sum_{k=1}^{N} a_k\, y(m-k) + \sum_{k=0}^{M} b_k\, x(m-k)$$

Figure shows a direct form implementation of equation (but with only output delays summation) an impulse, the output pe known as an *Infinite Dur* for an IIR filter include fe regressive-moving-averag signal processing literatu



Point to remember :

Output of the IIR Filter is dependent on previous input values AND previous output values

FINITE IMPULSE RESPONSE (FIR) FILTERS

(ALSO KNOWN AS NON-RECURSIVE FILTER)

A FIR filter (Finite Impulse Response) which is also known as a non-recursive filter has a simpler transfer function which does not contain any denominator terms.

$$H(z) = \frac{y_n(z)}{x_n(z)} = \frac{(b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots.. + b_M z^{-M})}{(a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots.. + a_N z^{-N})}$$

The coefficient a_0 is usually taken to be equal to 1, and all the other a coefficients are zero. The transfer function of a second-order FIR filter can therefore be expressed in the general form

$$\frac{y_n(z)}{x_n(z)} = b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots.. + b_M z^{-M}$$

Transfer Function (z-domain)

$$y(m) = \sum_{k=0}^{M} b_k \, x(m-k)$$

Difference Equation (discrete time samples)

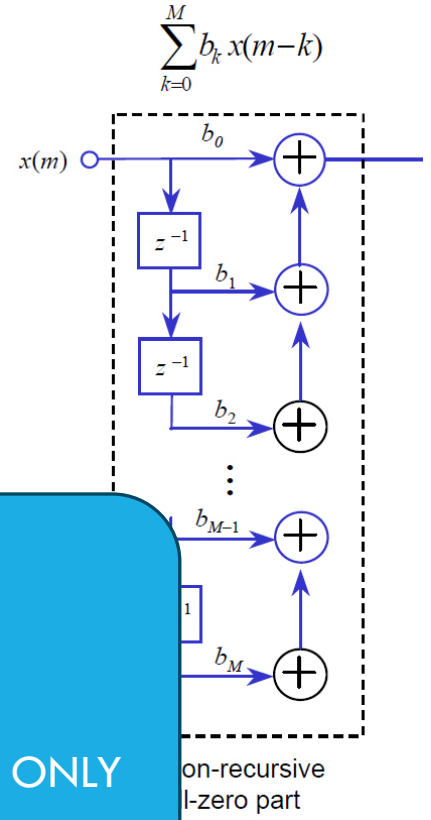
FINITE IMPULSE RESPONSE (FIR) FILTERS

(ALSO KNOWN AS NON-RECURSIVE FILTER)

A non-recursive filter has no feedback, and its input-output relation is given by :

$$y(m) = \sum_{k=0}^{M} b_k \, x(m-k)$$

As shown in equation above, the output $y(m)$ of a non-recursive filter is a function only of the input signal $x(m)$. The response of such a filter to an impulse consists of a finite sequence of $M+1$ sa
known as a *Finite-Duratio*
non-recursive filter includ
(MA) filter a term usually



Point to remember :

Output of the FIR Filter is dependent on previous input values ONLY

HOW YOU USE/IMPLEMENT DIGITAL FILTERS?

FIR FILTERS (SIMPLE DIGITAL FILTER)

non-recursive Filter

Three-Term Average Filter :

$$y_n = \frac{x_n + x_{n-1} + x_{n-2}}{3}$$

The output is the average (arithmetic mean) of the current and previous input :

$$y_0 = \frac{x_0 + x_{-1} + x_{-2}}{3}$$

$$y_1 = \frac{x_1 + x_0 + x_{-1}}{3}$$

$$y_2 = \frac{x_2 + x_1 + x_0}{3}$$

$$y_3 = \frac{x_3 + x_2 + x_1}{3}$$

$$\ldots etc$$

As before, x_{-1} and x_{-2} are taken to be zero.

FIR FILTERS (SIMPLE DIGITAL FILTER)

Three-Term Average Filter :

$$y_n = \frac{x_n + x_{n-1} + x_{n-2}}{3}$$

The output is the average (arithmeti

As before, x_{-1} and x_{-2} are taken to be ze

Discrete Difference Equation :

$$V_{out_n} = \frac{1}{3} V_{in_n} + \frac{1}{3} V_{in_{n-1}} + \frac{1}{3} V_{in_{n-2}}$$

$$V_{out}[n] = \frac{1}{3} V_{in}[n] + \frac{1}{3} V_{in}[n-1] + \frac{1}{3} V_{in}[n-2]$$

FIR FILTERS (SIMPLE DIGITAL FILTER)

Central Difference Filter :

$$y_n = \frac{x_n - x_{n-2}}{2}$$

non-recursive Filter

This is similar in its effect to example (4). The output is equal to half the change in the input signal over the previous two sampling intervals:

$$y_0 = \frac{x_0 - x_{-2}}{2}$$
$$y_1 = \frac{x_1 - x_{-1}}{2}$$
$$y_2 = \frac{x_2 - x_0}{2}$$
$$y_3 = \frac{x_3 - x_1}{2}$$
$$\ldots etc$$

FIR FILTERS (DIGITAL FILTER DESIGN)

A digital filter has been introduced in the earlier slides and we can express their respective equations into the corresponding difference equation (input-output equation) and their respective Transfer Function in z-domain.

For the averaging filter, we can easily tell that the filter type is a Low-Pass Digital Filter but it is not so easy to tell the Filter type from the equation. What is also lacking at the moment is that we have not customised the averaging filter to any specific cut-off frequency nor sampling rate.

Designing proper FIR filters via windowing methods will requires some signal processing understanding and this is beyond the scope of this year2 course.

IIR FILTERS (DIGITAL FILTER DESIGN)

What we will cover for this Year2 course is a method for designing IIR Filters (Infinite Impulse Response). This is the Method of Using Bilinear Transformation on existing Analogue Filter Responses to convert them into a IIR Digital Filter Implementation.

DIGITAL FILTER DESIGN STEPS (IIR)

1. Decide on the Filter Type
2. Design an Analogue Filter
3. Digitize the Analogue Filter using the Bilinear Transform
4. Implement the Digital Filter

First 2 steps are the same for Analogue Filter Design

COMPLETE CYCLE :

ANALOGUE DOMAIN

LPF 1st Order Differential Eq (time)

$$V_i(t) = RC\frac{dV_o(t)}{dt} + V_o(t)$$



LPF Transfer Function (Laplace)

$$H_{LPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{1/RC}{s + 1/RC} = \frac{\omega_c}{s + \omega_c}$$

DIGITAL DOMAIN

LPF Difference Eq (discrete/sampled time)

$$y(n) = 0.969y(n-1) + 0.0155x(n) + 0.0155x(n-1)$$



LPF Transfer Function (Z-Domain)

$$H(z) = \frac{V_o(z)}{V_i(z)} = \frac{0.0155 + 0.0155z^{-1}}{1 - 0.969z^{-1}}$$

Bilinear Transformation

EX DIGITAL FILTER DESIGN

Given the standard 1st Order Analogue Low-Pass Analogue Filter (Laplace Transform Domain). Derive the Coefficients of the Low-Pass Digital Filter (IIR) using the Bilinear Transformation method.

You are to design using a corner frequency of 5 Hz and a very generous sampling frequency of 1 KHz.

$$2\pi f_c = \frac{1}{RC} \quad \text{or} \quad \omega_c = \frac{1}{RC}$$

$$H_{LPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{\omega_c}{s + \omega_c}$$

Low-Pass Filter in the Laplace Transform Domain (derived earlier)

ANALOGUE TO DIGITAL FILTER CONVERSION

(BILINEAR TRANSFORMATION)

Converting from Laplace Transform Domain (analogue) to Z-Transform Domain (digital)

$$H_{LPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{{}^{1}/_{RC}}{s+{}^{1}/_{RC}} = \frac{\omega_c}{s+\omega_c}$$

Given :

Sampling frequency : $f_s = 1KHz$

Sampling period : $\delta t = \frac{1}{f_s} = 1ms$

Bilinear transform :

Let $s = \frac{2}{\delta t}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$ or $\frac{2}{\delta t}\left(\frac{z-1}{z+1}\right)$

$$H_{LPF}(z) = \frac{2\pi(5)}{\frac{2}{\delta t}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 2\pi(5)}$$

ANALOGUE TO DIGITAL FILTER CONVERSION

(BILINEAR TRANSFORMATION)

$$H_{LPF}(z) = \frac{2\pi(5)}{\frac{2}{\delta t}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 2\pi(5)} \qquad = \frac{2\pi(5)}{2000\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 2\pi(5)}$$

Transfer Function of the Low-Pass Filter in the Z-Transform Domain (digital)

difference equation (you can implement in your software)

ANALOGUE TO DIGITAL FILTER CONVERSION

(BILINE

H_{LI}

$$\frac{V_o(z)}{V_i(z)} = \frac{0.0155 + 0.0155z^{-1}}{1 - 0.969z^{-1}}$$

$$V_o(z)(1 - 0.969z^{-1}) = V_i(z)(0.0155 + 0.0155z^{-1})$$

$$V_o(z) - 0.969z^{-1}(V_o(z)) = (0.0155)V_i(z) + (0.0155)z^{-1}(V_i(z))$$

$$V_o(z) = 0.969z^{-1}\big(V_o(z)\big) + (0.0155)V_i(z) + (0.0155)z^{-1}(V_i(z))$$

Convert the Z-Transform Equation into a Discrete Equation :

$$y(n) = 0.969y(n-1) + 0.0155x(n) + 0.0155x(n-1)$$

z^{-1} can be considered as a delay operator

Discrete Difference Equation you can implement via software

EX DIGITAL FILTER DESIGN

Given the standard 1st Order Analogue Low-Pass Analogue Filter (Laplace Transform Domain). Derive the Coefficients of the Low-Pass Digital Filter (IIR) using the Bilinear Transformation method.

You are to design using a corner frequency of **10** Hz and a very generous sampling frequency of 1 KHz.

$$2\pi f_c = \frac{1}{RC} \quad \text{or} \quad \omega_c = \frac{1}{RC}$$

$$H_{LPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{\omega_c}{s + \omega_c}$$

Low-Pass Filter in the Laplace Transform Domain (derived earlier)

ANALOGUE TO DIGITAL F… …RSION

Another Example with $f_c = 10\ Hz$

$$H(z) = \frac{2\pi(10)}{\frac{2}{\delta t}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 2\pi(10)} = \frac{}{2000\left(\frac{1}{1+z}\right.}$$

$$\frac{V_o(z)}{V_i(z)} = \frac{2\pi(10)(1+z^{-1})}{2000(1-z^{-1}) + 2\pi(10)(1+z^{-1})}$$

$$\frac{V_o(z)}{V_i(z)} = \frac{20\pi + 20\pi z^{-1}}{2000 + 20\pi + (-2000 + 20\pi)z^{-1}} = \frac{62.832 + 62.}{2062.832 - 1937.168}$$

$$\frac{V_o(z)}{V_i(z)} = \frac{0.0305 + 0.0305z^{-1}}{1 - 0.939z^{-1}}$$

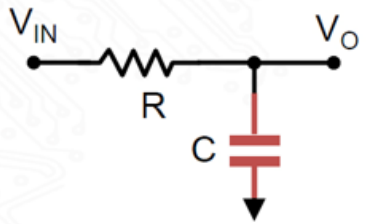
$$y(n) - 0.939y(n-1) = 0.0305x(n) + 0.0305x(n-1)$$

$$y(n) = 0.939y(n-1) + 0.0305x(n) + 0.0305x(n-1)$$

Comparison of Digital Filter Design(Bilinear Transformation) with the Backward Euler Approximate Method (Numerical Method for 1st Order ODE)

ANALOGUE TO DIGITAL F… …RSION

Digital Low Pass Filtering



$V_O + RC\, dV_O/dt = V_{IN}$ ------ (1)

$dV_O/dt = (V_O[n] - V_O[n-1]) / T$ ------ (2)

Substitute (2) into (1) and rearranging:

$V_O[n] = T/(T+RC)\, V_{IN}[n] + RC/(T+RC)\, V_O[n-1]$

Typical LPF: $T/(T+RC) << RC/(T+RC)$

- E.g. $V_O[n] = 0.1V_{IN}[n] + 0.9\, V_O[n-1]$

(More in-depth derivation later in the topic on Digital Filter)

numerical method for solving 1st Order Differential Equation

Backward Euler Method

This is a difference equation (discrete time equation) which you can implement on software

Comp… …ear … 1st

$$y(n) = 0.939y(n-1) + 0.0305x(n) + 0.0305x(n-1)$$

ANALOGUE TO

(BILINEAR TRANSFORMATIO

$$H(z) = \frac{2\pi(10)}{\frac{2}{\delta t}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 2\pi(1}$$

$$\frac{V_o(z)}{V_i(z)} = \frac{}{2}$$

$$\frac{V_o(z)}{V_i(z)} = \frac{}{2}$$

$$\frac{V_o(z)}{V_i(z)} = \frac{}{1 - 0.939z^{-1}}$$

Answer T2Q2i:

Find the coefficients value of the digital filter

$$V_o[n] = \left(\frac{\Delta t}{\Delta t+RC}\right)V_{in}[n] + \left(\frac{RC}{\Delta t+RC}\right)V_o[n-1]$$

(i) Bandwidth of the filter = 10 Hz
Time interval in between samples computation (i.e. [n] and [n-1]) Δt = 1msec)
Bandwidth of the filter $1/(2\pi RC)$ = 10Hz

$$RC = 0.016,\ \Delta t = 0.001$$

$$\left(\frac{\Delta t}{\Delta t+RC}\right) = 0.06$$

$$\left(\frac{RC}{\Delta t+RC}\right) = 0.94$$

$$V_o[n] = (0.06)V_{in}[n] + (0.94)V_o[n-1]$$

$$y(n) - 0.939y(n-1) = 0.0305x(n) + 0.0305x(n-1)$$

$$y(n) = 0.939y(n-1) + 0.0305x(n) + 0.0305x(n-1)$$

EX DIGITAL FILTER DESIGN

HOMEWORK

Given the standard 1st Order Analogue High-Pass Analogue Filter (Laplace Transform Domain). Derive the Coefficients of the High-Pass Digital Filter (IIR) using the Bilinear Transformation method.

You are to design using a corner frequency of 5 Hz and a very generous sampling frequency of 1 KHz.

$$H_{HPF}(s) = \frac{V_o(s)}{V_i(s)} = \frac{s}{s + \omega_c}$$

SUMMARY

Analogue Filters (Type

1st Order Low-Pass/Hig

Design of Analogue Filt

Introduction to Digital F

Simple Digital Filters

Design of Digital Filter (Bilinear Transformation)

LEARNING OBJECTIVES :

Derive the Transfer Function of Analogue Low-Pass/High-Pass Filters. in the Time Domain, Frequency Domain, Laplace Transform Domain

Design considerations when using an Analogue Filter.

Digital Filters – Introduction

Digital Filter Design Method (via Bilinear Transformation)

WATCH A VIDEO

How to design and implement a digital low-pass filter on an Arduino

QUICK SUMMARY (FILTER)

$f_c = 10Hz,\ \ sampling\ freq = 1KHz$

ANALOGUE DOMAIN

LPF 1st Order Differential Eq (time)

$$V_i(t) = RC\frac{dV_o(t)}{dt} + V_o(t)$$



LPF Transfer Function (Laplace)

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1/RC}{s + 1/RC} = \frac{\omega_c}{s + \omega_c}$$

DIGITAL DOMAIN

LPF Difference Eq (discrete/sampled time)

$$y(n) = 0.969y(n-1) + 0.0155x(n) + 0.0155x(n-1)$$



LPF Transfer Function (Z-Domain)

$$H(z) = \frac{V_o(z)}{V_i(z)} = \frac{0.0155 + 0.0155z^{-1}}{1 - 0.969z^{-1}}$$

LOOKING AHEAD (CONTROL)

ANALOGUE DOMAIN

N[th] Order Differential Eq (time)

$$V_{i(t)} = x\frac{dV_{o(t)}}{dt} + wV_o(t)$$

$$V_{i(t)} = a\frac{d^2V_{o(t)}}{dt^2} + b\frac{dV_{o(t)}}{dt} + cV_o(t)$$



System Transfer Function (Laplace)

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{a}{s+b}$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{d}{(s+e)(s+f)} = \frac{d}{s^2 + (e+f)s + ef}$$

DIGITAL DOMAIN

System Difference Eq (discrete/sampled time)

$$y(n) = ay(n-1) + bx(n) + cx(n-1) + \cdots.$$

$$y(n) = ey(n-1) + fy(n-2) + gx(n) + hx(n-1) + \ldots$$



LPF Transfer Function (Z-Domain)

$$H(z) = \frac{V_o(z)}{V_i(z)} = \frac{a + bz^{-1}}{1 + cz^{-1}}$$

$$H(z) = \frac{V_o(z)}{V_i(z)} = \frac{e + fz^{-1} + \cdots}{1 + gz^{-1} + hz^{-2} + \ldots}$$

DIFFERENCE EQUATION – Z TRANSFORM TRANSFER FUNCTION

ANALOGUE DOMAIN

LPF 1st Order Differential Eq (time)

$$V_i(t) = RC\frac{dV_o(t)}{dt} + V_o(t)$$



LPF Transfer Function (Laplace)

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1/RC}{s + 1/RC} = \frac{\omega_c}{s + \omega_c}$$

DIGITAL DOMAIN

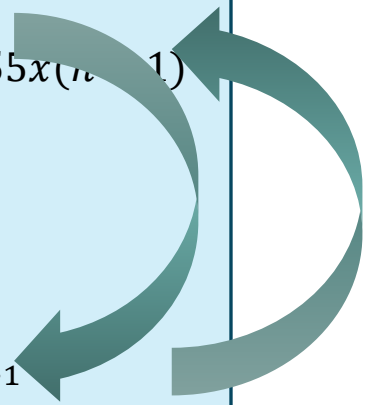
LPF Difference Eq (discrete/sampled time)

$$y(n) = 0.969y(n-1) + 0.0155x(n) + 0.0155x(n-1)$$



LPF Transfer Function (Z-Domain)

$$H(z) = \frac{V_o(z)}{V_i(z)} = \frac{0.0155 + 0.0155z^{-1}}{1 - 0.969z^{-1}}$$



DIFFERENCE EQUATION – Z TRANSFORM TRANSFER FUNCTION

The operator z^{-1} to an input value (say x_n) gives the previous input (x_{n-1}) :

$$z^{-1}x_n = x_{n-1}$$

Similarly, you can apply the operator z^{-1} to the output value

$$z^{-1}y_n = y_{n-1}$$

Applying the delay operator z^{-1} twice produces a delay of two sampling intervals :

$$z^{-1}(z^{-1}x_n) = z^{-1}x_{n-1} = x_{n-2}$$

$$z^{-2}x_n = x_{n-2}\ , \quad z^{-2}y_n = y_{n-2}$$

DIFFERENCE EQUATION – Z TRANSFORM TRANSFER FUNCTION

For example, using the three-term average filter, the difference equation is :

$$y_n = \frac{x_n + x_{n-1} + x_{n-2}}{3} \quad \text{or} \quad y(n) = \frac{x(n)+x(n-1)+)+x(n-2)}{3}$$

Difference Equation can be written in either form

Introducing the z^{-1} operator :

$$y_n = \frac{x_n + x_n z^{-1} + x_n z^{-2}}{3}$$

The transfer function for the filter is therefore :

$$y_n(z) = \frac{(1 + z^{-1} + z^{-2})x_n(z)}{3}$$

Z Transform
Transfer function

$$\frac{y_n(z)}{x_n(z)} = \frac{(1+z^{-1}+z^{-2})}{3} = \frac{(z^2+z^1+1)}{3z^2}$$

Z TRANSFORM TRANSFER FUNCTION – DIFFERENCE EQUATION

$$H(z) = \frac{y_n(z)}{x_n(z)} = \frac{(b_0 + b_1 z^{-1} + b_2 z^{-2})}{(a_0 + a_1 z^{-1} + a_2 z^{-2})}$$

Z Transform
Transfer function

Replacing the numerator/dominator :

$$y_n(z)\,(a_0 + a_1 z^{-1} + a_2 z^{-2}) = x_n(z)(b_0 + b_1 z^{-1} + b_2 z^{-2})$$

$$a_0\, y_n(z) + a_1\, z^{-1}\, y_n(z) + a_2 z^{-2}\, y_n(z) = b_0\, x_n(z) + b_1 z^{-1}\, x_n(z) + b_2 z^{-2}\, x_n(z)$$

Since we know z^{-1} is a delay operator (of one sample) :

$$a_0\, y(n) + a_1\, y(n-1) + a_2\, y(n-2) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2)$$

$$a_0\, y(n) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) - a_1 y(n-1) - a_2\, y(n-2)$$

Difference Equation
(discrete time samples)

also known as
Input – Output Equation